

CONFIDENTIAL



UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION
SEMESTER II 2022/2023

SUBJECT CODE : SCS/SECI 1143/2143
SUBJECT NAME : PROBABILITY & STATISTICAL DATA ANALYSIS
YEAR/COURSE :
TIME :
DATE :
VENUE :

INSTRUCTIONS TO THE STUDENTS:

Answer all questions in the answer booklet.

Do all calculations in 3 decimal points.

NAME	
MATRIC NO.	
SECTION	
LECTURER	

(This question paper consist of 13 pages including this pages)

QUESTION 1

[10 MARKS]

- a) A recent study investigated the usage of ChatGPT towards the students. ChatGPT is a tool that provide language learning, translation and many more. Researcher observed that a representative sample of students from Faculty of Computing found that 158 out of 367 students are active using the ChatGPT.
- i) Compute a point estimate of the true proportion of all students who active using the ChatGPT.
(1 mark)
 - ii) Construct and interpret a 95% confidence interval for the true proportion of all students who actively using ChatGPT services.
(2 marks)
- b) In a study of coffee lover among staff in UTM, 30 randomly selected staffs were asked how much time they drink coffee in a day. The time is assumed normally distributed and the sample standard deviation was 6 hours. At $\alpha = 0.01$, is there enough evidence to reject the claim that the population variance is at 35 hours.
- i) State the null and alternative hypothesis.
(1 mark)
 - ii) Calculate the test statistic value.
(2 marks)
 - iii) Find the critical value.
(1 mark)
 - iv) Draw and shade the critical region (rejection area) and what can you conclude from the test.
(2 marks)

QUESTION 2**[20 MARKS]**

- a) A research study was conducted to examine the differences between older and younger adults on perceived life satisfaction. A pilot study was conducted to examine this hypothesis. 13 older adults (over the age of 70) and 16 younger adults (between 20 and 30) were given a life satisfaction test. The summary of the score results for the test is given in Table 1.

Older adults	Younger adults
$s_1=1.21$	$s_2=0.63$

Table 1: Data on perceived life satisfaction

Test at the $\alpha = 0.05$ level of significance whether there is a difference in the variability between the two groups?

(10 marks)

- b) The iron contents of fruits before and after applying farm yard manure were observed as in Table 2.

Table 2: The iron contents of fruits before and after applying farm yard manure

Fruit No	1	2	3	4	5	6	7	8	9
Before Applying	11.6	18.7	15.9	20.6	10.1	17.4	7.2	12.2	11.7
After Applying	10.0	21.6	13.9	22.8	11.5	16.2	8.1	10.8	9.6

Is there any significant difference between the mean iron contents in the fruits before and after the farm yard manure at $\alpha = 0.05$ level of significance?

(10 marks)

QUESTION 3**[20 MARKS]**

- a) A staff member of an emergency medical service wishes to determine whether the number of accidents is equally distributed during a week. A week was selected at random, and the following data (Table 3) were obtained. Is there evidence to reject the hypothesis that the number of accident is equally distributed throughout the week, at 95% confidence level.

Table 3: Distribution number of accident during a week

Day	Mon.	Tues.	Wed.	Thurs.	Fri.	Sat.	Sun.
No. of accident	28	21	15	14	32	23	19

(8 marks)

- b) A study is being conducted to determine whether there is relationship between jogging and blood pressure. A random sample of 210 subjects is selected, and the raw data is shown in Table 4.

Table 4: Data from 210 respondents

Jogging Status	Blood pressure level	Count
Joggers	Low	34
Non joggers	Low	15
Joggers	Moderate	57
Non joggers	Moderate	63
Joggers	High	21
Non joggers	High	20

Based on the data,

- i) Construct the two ways contingency table based on data in Table 4.

(2 marks)

- ii) At $\alpha = 0.05$ level of significance, test the claim that jogging and blood pressure are not related.

(10 marks)

QUESTION 4**[15 MARKS]**

A study of the amount of rainfall and the quantity of air pollution removed produced the following data (Table 5). Assume that x and y are random variables with a bivariate normal distribution.

Table 5: Rainfall and air pollution data

Daily rainfall, x (0.01 cm)	Particulate removed, y ($\frac{\mu g}{m^3}$)
4.3	126
4.5	121
5.9	116
5.6	118
6.1	114
5.2	118
3.8	132
2.1	141
7.5	108

a) Draw (sketch) the scatter plot.

(2 marks)

b) Given that $\sum x^2 = 244.26$, $\sum y^2 = 133786$, calculate r .

(10 marks)

c) Does the value of r value found in 4(b) consistent with the pattern of the scatter plot in 4(a)?

(1 mark)

d) Is the r value found in 4(b) suggest a strong linear relationship? If yes, do the changes in x and y will go in the same direction?

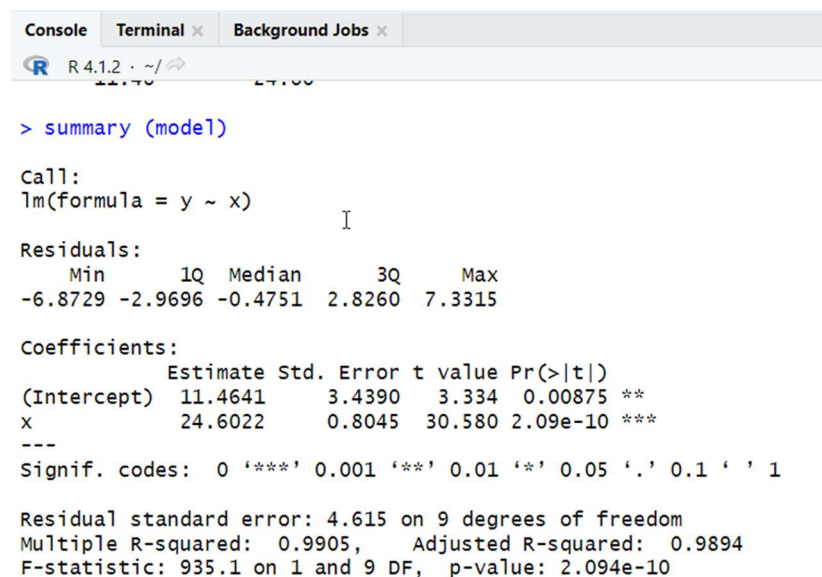
(2 marks)

QUESTION 5**[15 MARKS]**

Accu-Copiers, Inc., sells and services the Accu-500 copying machine. As part of its standard service contract, the company agrees to perform routine service on this copier. To obtain information about the time it takes to perform routine service, Accu-Copiers has collected data for 11 service calls. The data are given in Table 6. Whilst, the R output from fitting a least squares regression line to the data is shown in Figure 1.

Table 6: The service time data

Service Call	Number of copiers serviced, x	Number of minutes required, y
1	4	109
2	2	58
3	5	138
4	7	189
5	1	37
6	3	82
7	4	103
8	5	134
9	2	68
10	4	112
11	6	154



```

Console Terminal x Background Jobs x
R 4.1.2 ~ /
> summary (model)

Call:
lm(formula = y ~ x)

Residuals:
    Min       1Q   Median       3Q      Max
-6.8729 -2.9696 -0.4751  2.8260  7.3315

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  11.4641     3.4390   3.334  0.00875 **
x            24.6022     0.8045  30.580 2.09e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.615 on 9 degrees of freedom
Multiple R-squared:  0.9905,    Adjusted R-squared:  0.9894
F-statistic: 935.1 on 1 and 9 DF,  p-value: 2.094e-10

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Figure 1: R Output

- a) Find the least squares point estimates b_0 and b_1 on the R output (Figure 1) and write their value.

(2 marks)

- b) Interpret b_0 and b_1 . Does the interpretation of b_0 make practical sense? (3 marks)
- c) Write the equation of the least squares line. (2 marks)
- d) Use the least square line to compute a point estimate of the mean time to service four copiers. (2 marks)
- e) How many percent of the variation in the number of minutes required to service is explained by the variation in the number of copiers serviced? (1 mark)
- f) Conduct the t -test to check whether there is a linear relationship between x and y . Use $\alpha = 0.05$ (5 marks)

QUESTION 6

[20 MARKS]

A scientist wanted to see if mice is sensitive to colour. He set up an experiment using mice and different door colours to see if there is any colour that the mice would be attracted to. Three sets of

five mice were randomly selected to be placed in a standard maze but with different colour doors: red, green, and black. The objective of the experiment is to identify the time for each mouse to run in the maze and reach the door. He recorded the response time (in minutes), to which the time required for each mouse to complete the maze (Table 7).

Table 7: Response time recorded for each mouse to complete the maze.

Door Colour	Response time (minutes)				
	M1	M2	M3	M4	M5
Red	9	11	10	9	15
Green	20	21	23	17	30
Black	6	5	8	14	7

- Find the mean, $\bar{x}$, and standard deviation, s , for each door colour.
(6 marks)
- Calculate the test statistics, F .
(10 marks)
- At the 0.01 level of significance, test if the data provide sufficient evidence to conclude that mice are sensitive to different colour.
(4 marks)

List of Formula:

- Point Estimate and CI

$$\hat{p} \pm (z \text{ critical value}) \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

(b) Hypothesis Testing for one sample

$$\text{Test Statistic, } z_0 = \frac{\underline{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$\text{Test Statistic, } t_0 = \frac{\underline{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$\text{Test Statistic, } \chi^2 = \frac{(n - 1)s^2}{\sigma^2}$$

(c) Hypothesis Testing for two samples

$$\text{Test Statistic, } z_0 = \frac{\underline{x}_1 - \underline{x}_2 - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$\text{Test Statistic, } t_0 = \frac{\underline{x}_1 - \underline{x}_2 - \Delta_0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}; s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$\text{Test Statistic, } T_0 = \frac{\underline{x}_1 - \underline{x}_2 - \Delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}; v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

$$\text{Test Statistic, } F_0 = \frac{s_1^2}{s_2^2}$$

$$\text{Mean of the differences, } \underline{D} = \frac{\sum D}{n}$$

$$\text{Standard deviation of the differences, } S_D = \sqrt{\frac{\sum D^2 - \frac{(\sum D)^2}{n}}{n-1}}$$

$$\text{Test Statistic, } t = \frac{\frac{\bar{D} - \mu_D}{\frac{S_D}{\sqrt{n}}}}{\sqrt{n}}; \text{dof} = n - 1$$

(d) Chi-Square Test

$$\text{Test Statistic, } \chi^2 = \sum_i^n \frac{(o_i - e_i)^2}{e_i}$$

$$\text{Degree of Freedom for two-way contingency table} = (r-1)(c-1)$$

(e) Correlation

$$\text{Pearson Correlation Coefficient, } r = \frac{\sum xy - \frac{(\sum x)(\sum y)}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)\left(\sum y^2 - \frac{(\sum y)^2}{n}\right)}}$$

$$\text{Spearman Correlation Coefficient, } r_s = 1 - \frac{6 \sum d_i^2}{n(n^2-1)} \text{ where } d_i = y_i - x_i$$

$$\text{Test Statistic, } t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$$

(f) Regression

$$b_1 = \frac{\sum xy - \frac{(\sum x)(\sum y)}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

(g) Analysis of variance (ANOVA)

$$\textit{Test Statistic}, F = \frac{ns^2_{\bar{x}}}{s_p^2}$$

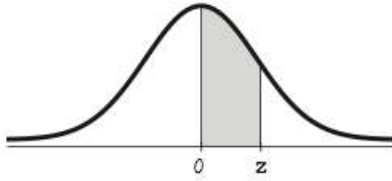


TABLE A-2 Standard Normal (z) Distribution

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	* .4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	↑ .4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	* .4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	↑ .4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.10 and higher	.4999									

NOTE: For values of z above 3.09, use 0.4999 for the area.

*Use these common values that result from interpolation:

z score	Area
1.645	0.4500
2.575	0.4950

From Frederick C. Mosteller and Robert E. K. Rourke, *Sturdy Statistics*, 1973, Addison-Wesley Publishing Co., Reading, MA. Reprinted with permission of Frederick Mosteller.

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.												
Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09		
-3.9	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003	.0003		
-3.8	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005		
-3.7	.0011	.0010	.0010	.0010	.0009	.0009	.0008	.0008	.0008	.0008		
-3.6	.0016	.0015	.0015	.0014	.0014	.0013	.0013	.0012	.0012	.0011		
-3.5	.0023	.0022	.0022	.0021	.0020	.0019	.0019	.0018	.0017	.0017		
-3.4	.0034	.0032	.0031	.0030	.0029	.0028	.0027	.0026	.0025	.0024		
-3.3	.0048	.0047	.0045	.0043	.0042	.0040	.0039	.0038	.0036	.0035		
-3.2	.0069	.0066	.0064	.0062	.0060	.0058	.0056	.0054	.0052	.0050		
-3.1	.0097	.0094	.0090	.0087	.0084	.0082	.0079	.0076	.0074	.0071		
-3.0	.0135	.0131	.0126	.0122	.0118	.0114	.0111	.0107	.0104	.0101		
-2.9	.0187	.0181	.0175	.0169	.0164	.0159	.0154	.0149	.0144	.0139		
-2.8	.0256	.0248	.0240	.0233	.0226	.0219	.0212	.0205	.0199	.0193		
-2.7	.0347	.0336	.0326	.0317	.0307	.0298	.0289	.0280	.0272	.0264		
-2.6	.0466	.0453	.0440	.0427	.0415	.0402	.0391	.0379	.0368	.0357		
-2.5	.0621	.0604	.0587	.0570	.0554	.0539	.0523	.0508	.0494	.0480		
-2.4	.0820	.0798	.0776	.0755	.0734	.0714	.0695	.0676	.0657	.0639		
-2.3	.1072	.1044	.1017	.0990	.0964	.0939	.0914	.0889	.0866	.0842		
-2.2	.1390	.1355	.1321	.1287	.1255	.1222	.1191	.1160	.1130	.1101		
-2.1	.1786	.1743	.1700	.1659	.1618	.1578	.1539	.1500	.1463	.1426		
-2.0	.2275	.2222	.2169	.2118	.2068	.2018	.0190	.0193	.0187	.0181		
-1.9	.2872	.2807	.2743	.2680	.2619	.2559	.2500	.2442	.2385	.2330		
-1.8	.3593	.3515	.3438	.3362	.3288	.3216	.3144	.3074	.3005	.2938		
-1.7	.4457	.4363	.4272	.4182	.4093	.4006	.3920	.3836	.3754	.3673		
-1.6	.5480	.5370	.5262	.5155	.5050	.4947	.4846	.4746	.4648	.4551		
-1.5	.6681	.6553	.6436	.6301	.6178	.6057	.5938	.5831	.5705	.5592		
-1.4	.8076	.7927	.7780	.7636	.7493	.7353	.7215	.7078	.6944	.6811		
-1.3	.9680	.9510	.9342	.9176	.9012	.8851	.8691	.8534	.8379	.8226		
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.9853		
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702		
-1.0	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786		
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109		
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673		
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476		
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510		
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760		
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207		
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827		
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591		
-0.1	.46017	.45620	.45224	.44828	.44433	.44038	.43644	.43251	.42858	.42465		
-0.0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414		

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.												
Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09		
0	.5000	.5039	.5078	.5117	.5155	.5194	.5232	.5270	.5318	.5366		
0.1	.5398	.5438	.5476	.5512	.5547	.5581	.5614	.5646	.5678	.5714		
0.2	.5796	.5831	.5870	.5906	.5943	.5971	.6027	.6064	.6102	.6149		
0.3	.6179	.6212	.6252	.6290	.6330	.6368	.6408	.6441	.6483	.6513		
0.4	.6544	.6590	.6626	.6664	.6703	.6736	.6774	.6802	.6839	.6873		
0.5	.6914	.6947	.6977	.7014	.7054	.7088	.7126	.7156	.7194	.7240		
0.6	.7257	.7290	.7327	.7365	.7391	.7421	.7457	.7487	.7515	.7549		
0.7	.7580	.7615	.7644	.7673	.7705	.7737	.7767	.7795	.7830	.7854		
0.8	.7881	.7913	.7938	.7963	.7985	.8024	.8051	.8078	.8107	.8137		
0.9	.8159	.8189	.8211	.8238	.8263	.8284	.8314	.8338	.8364	.8391		
1.0	.8413	.8437	.8461	.8484	.8503	.8514	.8543	.8569	.8593	.8614		
1.1	.8643	.8660	.8684	.8706	.8726	.8743	.8768	.8790	.8810	.8828		
1.2	.8849	.8866	.8877	.8905	.8921	.8945	.8967	.8976	.8997	.9017		
1.3	.9030	.9049	.9068	.9084	.9098	.9119	.9130	.9146	.9161	.9174		
1.4	.9194	.9207	.9220	.9234	.9247	.9257	.9278	.9292	.9306	.9319		
1.5	.9331	.9348	.9374	.9399	.9422	.9443	.9462	.9479	.9495	.9508		
1.6	.9520	.9530	.9548	.9584	.9590	.9608	.9614	.9624	.9637	.9649		
1.7	.9543	.9567	.9578	.9588	.9597	.9608	.9614	.9624	.9637	.9649		
1.8	.9640	.9648	.9652	.9658	.9671	.9678	.9686	.9696	.9702	.9712		
1.9	.9712	.9719	.9727	.9730	.9731	.9741	.9750	.9758	.9761	.9770		
2.0	.9775	.9778	.9781	.9782	.9792	.9792	.9800	.9807	.9814	.9819		
2.1	.9821	.9825	.9830	.9841	.9832	.9842	.9846	.9850	.9857	.9874		
2.2	.9861	.9865	.9879	.9871	.9874	.9878	.9889	.9890	.9894	.9899		
2.3	.9928	.9956	.9983	.9901	.9906	.9921	.9941	.9946	.9951	.9958		
2.4	.9910	.9902	.9924	.9924	.9924	.9924	.9924	.9924	.9924	.9924		
2.5	.9937	.9936	.9943	.9943	.9946	.9946	.9946	.9946	.9946	.9946		
2.6	.9934	.9947	.9960	.9973	.9983	.9983	.9983	.9983	.9983	.9983		
2.7	.9963	.9964	.9964	.9964	.9963	.9963	.9963	.9963	.9963	.9963		
2.8	.9974	.9975	.9975	.9975	.9974	.9974	.9974	.9974	.9974	.9974		
2.9	.9981	.9981	.9981	.9981	.9981	.9981	.9981	.9981	.9981	.9981		
3.0	.9985	.9986	.9987	.9987	.9988	.9988	.9988	.9988	.9988	.9988		
3.1	.9990	.9990	.9991	.9991	.9991	.9991	.9991	.9991	.9991	.9991		
3.2	.9991	.9991	.9991	.9991	.9991	.9991	.9991	.9991	.9991	.9991		
3.3	.9992	.9993	.9993	.9993	.9993	.9993	.9993	.9993	.9993	.9993		
3.4	.9996	.9996	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997		
3.5	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997		
3.6	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998		
3.7	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999		
3.8	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999		
3.9	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999		

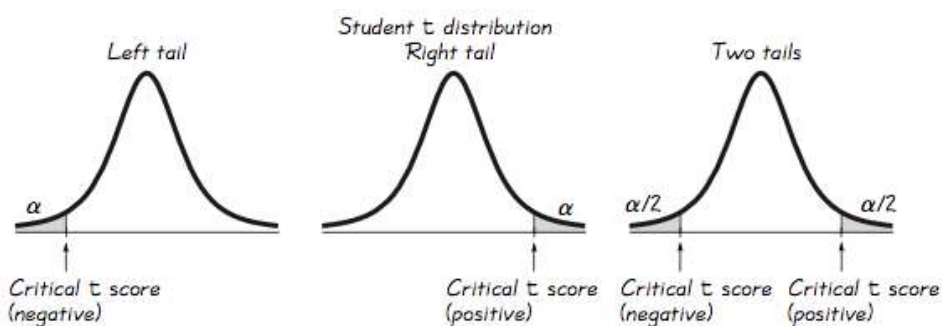


TABLE A-3 t Distribution

α						
Degrees of Freedom	.005 (one tail) .01 (two tails)	.01 (one tail) .02 (two tails)	.025 (one tail) .05 (two tails)	.05 (one tail) .10 (two tails)	.10 (one tail) .20 (two tails)	.25 (one tail) .50 (two tails)
1	63.657	31.821	12.706	6.314	3.078	1.000
2	9.925	6.965	4.303	2.920	1.886	.816
3	5.841	4.541	3.182	2.353	1.638	.765
4	4.604	3.747	2.776	2.132	1.533	.741
5	4.032	3.365	2.571	2.015	1.476	.727
6	3.707	3.143	2.447	1.943	1.440	.718
7	3.500	2.998	2.365	1.895	1.415	.711
8	3.355	2.896	2.306	1.860	1.397	.706
9	3.250	2.821	2.262	1.833	1.383	.703
10	3.169	2.764	2.228	1.812	1.372	.700
11	3.106	2.718	2.201	1.796	1.363	.697
12	3.054	2.681	2.179	1.782	1.356	.696
13	3.012	2.650	2.160	1.771	1.350	.694
14	2.977	2.625	2.145	1.761	1.345	.692
15	2.947	2.602	2.132	1.753	1.341	.691
16	2.921	2.584	2.120	1.746	1.337	.690
17	2.898	2.567	2.110	1.740	1.333	.689
18	2.878	2.552	2.101	1.734	1.330	.688
19	2.861	2.540	2.093	1.729	1.328	.688
20	2.845	2.528	2.086	1.725	1.325	.687
21	2.831	2.518	2.080	1.721	1.323	.686
22	2.819	2.508	2.074	1.717	1.321	.686
23	2.807	2.500	2.069	1.714	1.320	.685
24	2.797	2.492	2.064	1.711	1.318	.685
25	2.787	2.485	2.060	1.708	1.316	.684
26	2.779	2.479	2.056	1.706	1.315	.684
27	2.771	2.473	2.052	1.703	1.314	.684
28	2.763	2.467	2.048	1.701	1.313	.683
29	2.756	2.462	2.045	1.699	1.311	.683
Large (z)	2.575	2.326	1.960	1.645	1.282	.675

TABLE A-4		Chi-Square (χ^2) Distribution								
		Area to the Right of the Critical Value								
Degrees of Freedom	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	—	—	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.071	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.299
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.042	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.194	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.257	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.954	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

TABLE E

F critical values (continued)

		Degrees of freedom in the numerator									
		1	2	3	4	5	6	7	8	9	
8	.050	3.46	3.11	2.93	2.81	2.73	2.67	2.63	2.59	2.56	
	.025	4.00	3.64	3.47	3.34	3.26	3.19	3.15	3.11	3.09	
	.010	4.57	4.20	4.03	3.90	3.82	3.75	3.70	3.66	3.63	
	.005	5.05	4.68	4.51	4.38	4.30	4.23	4.18	4.14	4.11	
	.001	5.95	5.57	5.40	5.27	5.19	5.12	5.07	5.03	5.00	
9	.050	3.36	3.01	2.83	2.71	2.63	2.57	2.53	2.49	2.46	
	.025	3.90	3.54	3.37	3.24	3.16	3.09	3.05	3.01	2.99	
	.010	4.47	4.10	3.93	3.80	3.72	3.65	3.60	3.56	3.53	
	.005	4.95	4.58	4.41	4.28	4.20	4.13	4.08	4.04	4.01	
	.001	5.85	5.47	5.30	5.17	5.09	5.02	4.97	4.93	4.90	
10	.050	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.38	2.35	
	.025	3.83	3.47	3.30	3.17	3.09	3.02	2.97	2.93	2.90	
	.010	4.40	4.03	3.86	3.73	3.65	3.58	3.53	3.49	3.46	
	.005	4.88	4.51	4.34	4.21	4.13	4.06	4.01	3.97	3.94	
	.001	5.78	5.40	5.23	5.10	5.02	4.95	4.90	4.86	4.83	
11	.050	3.23	2.86	2.66	2.54	2.45	2.39	2.34	2.30	2.27	
	.025	3.77	3.41	3.24	3.11	3.03	2.96	2.91	2.87	2.84	
	.010	4.34	3.97	3.80	3.67	3.59	3.52	3.47	3.43	3.40	
	.005	4.82	4.45	4.28	4.15	4.07	4.00	3.95	3.91	3.88	
	.001	5.72	5.34	5.17	5.04	4.96	4.89	4.84	4.80	4.77	
12	.050	3.18	2.81	2.61	2.49	2.39	2.33	2.28	2.24	2.21	
	.025	3.72	3.36	3.19	3.06	2.98	2.91	2.86	2.82	2.79	
	.010	4.29	3.92	3.75	3.62	3.54	3.47	3.42	3.38	3.35	
	.005	4.77	4.40	4.23	4.10	4.02	3.95	3.90	3.86	3.83	
	.001	5.67	5.29	5.12	4.99	4.91	4.84	4.79	4.75	4.72	
13	.050	3.14	2.76	2.56	2.43	2.33	2.27	2.22	2.18	2.15	
	.025	3.68	3.32	3.15	3.02	2.94	2.87	2.82	2.78	2.75	
	.010	4.25	3.88	3.71	3.58	3.50	3.43	3.38	3.34	3.31	
	.005	4.73	4.36	4.19	4.06	3.98	3.91	3.86	3.82	3.79	
	.001	5.63	5.25	5.08	4.95	4.87	4.80	4.75	4.71	4.68	
14	.050	3.10	2.73	2.52	2.39	2.31	2.24	2.19	2.15	2.12	
	.025	3.64	3.28	3.11	2.98	2.90	2.83	2.78	2.74	2.71	
	.010	4.21	3.84	3.67	3.54	3.46	3.39	3.34	3.30	3.27	
	.005	4.69	4.32	4.15	4.02	3.94	3.87	3.82	3.78	3.75	
	.001	5.59	5.21	5.04	4.91	4.83	4.76	4.71	4.67	4.64	
15	.050	3.07	2.70	2.49	2.36	2.27	2.21	2.16	2.12	2.09	
	.025	3.61	3.25	3.08	2.95	2.87	2.80	2.75	2.71	2.68	
	.010	4.18	3.81	3.64	3.51	3.43	3.36	3.31	3.27	3.24	
	.005	4.66	4.29	4.12	3.99	3.91	3.84	3.79	3.75	3.72	
	.001	5.56	5.18	5.01	4.88	4.80	4.73	4.68	4.64	4.61	
16	.050	3.05	2.67	2.46	2.33	2.24	2.18	2.13	2.09	2.06	
	.025	3.59	3.23	3.06	2.93	2.85	2.78	2.73	2.69	2.66	
	.010	4.16	3.79	3.62	3.49	3.41	3.34	3.29	3.25	3.22	
	.005	4.64	4.27	4.10	3.97	3.89	3.82	3.77	3.73	3.70	
	.001	5.54	5.16	4.99	4.86	4.78	4.71	4.66	4.62	4.59	
17	.050	3.03	2.65	2.44	2.31	2.22	2.16	2.11	2.07	2.04	
	.025	3.57	3.21	3.04	2.91	2.83	2.76	2.71	2.67	2.64	
	.010	4.14	3.77	3.60	3.47	3.39	3.32	3.27	3.23	3.20	
	.005	4.62	4.25	4.08	3.95	3.87	3.80	3.75	3.71	3.68	
	.001	5.52	5.14	4.97	4.84	4.76	4.69	4.64	4.60	4.57	

Degrees of freedom in the denominator

TABLE E

F critical values (continued)

		Degrees of freedom in the numerator									
		10	12	15	20	25	30	40	50	60	100
11	.050	2.54	2.50	2.46	2.42	2.40	2.38	2.36	2.35	2.34	2.33
	.025	3.15	3.12	3.08	3.04	3.02	3.00	2.98	2.97	2.96	2.95
	.010	3.60	3.57	3.53	3.49	3.47	3.45	3.43	3.42	3.41	3.40
	.005	4.00	3.97	3.93	3.89	3.87	3.85	3.83	3.82	3.81	3.80
	.001	4.78	4.74	4.70	4.66	4.64	4.62	4.60	4.59	4.58	4.57
12	.050	2.52	2.48	2.44	2.40	2.37	2.35	2.33	2.32	2.31	2.30
	.025	3.13	3.10	3.06	3.02	3.00	2.98	2.96	2.95	2.94	2.93
	.010	3.58	3.55	3.51	3.47	3.45	3.43	3.41	3.40	3.39	3.38
	.005	3.98	3.95	3.91	3.87	3.85	3.83	3.81	3.80	3.79	3.78
	.001	4.76	4.72	4.68	4.64	4.62	4.60	4.58	4.57	4.56	4.55
13	.050	2.50	2.46	2.42	2.38	2.35	2.33	2.31	2.30	2.29	2.28
	.025	3.11	3.08	3.04	3.00	2.98	2.96	2.94	2.93	2.92	2.91
	.010	3.56	3.53	3.49	3.45	3.43	3.41	3.39	3.38	3.37	3.36
	.005	3.96	3.93	3.89	3.85	3.83	3.81	3.79	3.78	3.77	3.76
	.001	4.74	4.70	4.66	4.62	4.60	4.58	4.56	4.55	4.54	4.53
14	.050	2.48	2.44	2.40	2.36	2.33	2.31	2.29	2.28	2.27	2.26
	.025	3.09	3.06	3.02	2.98	2.96	2.94	2.92	2.91	2.90	2.89
	.010	3.54	3.51	3.47	3.43	3.41	3.39	3.37	3.36	3.35	3.34
	.005	3.94	3.91	3.87	3.83	3.81	3.79	3.77	3.76	3.75	3.74
	.001	4.72	4.68	4.64	4.60	4.58	4.56	4.54	4.53	4.52	4.51
15	.050	2.46	2.42	2.38	2.34	2.31	2.29	2.27	2.26	2.25	2.24
	.025	3.07	3.04	3.00	2.96	2.94	2.92	2.90	2.89	2.88	2.87
	.010	3.52	3.49	3.45	3.41	3.39	3.37	3.35	3.34	3.33	3.32
	.005	3.92	3.89	3.85	3.81	3.79	3.77	3.75	3.74	3.73	3.72
	.001	4.70	4.66	4.62	4.58	4.56	4.54	4.52	4.51	4.50	4.49
16	.050	2.44	2.40	2.36	2.32	2.29	2.27	2.25	2.24	2.23	2.22
	.025	3.05	3.02	2.98	2.94	2.92	2.90	2.88	2.87	2.86	2.85
	.010	3.50	3.47	3.43	3.39	3.37	3.35	3.33	3.32	3.31	3.30
	.005	3.90	3.87	3.83	3.79	3.77	3.75	3.73	3.72	3.71	3.70
	.001	4.68	4.64	4.60	4.56	4.54	4.52	4.50	4.49	4.48	4.47
17	.050	2.42	2.38	2.34	2.30	2.27	2.25	2.23	2.22	2.21	2.20
	.025	3.03	3.00	2.96	2.92	2.90	2.88	2.86	2.85	2.84	2.83
	.010	3.48	3.45	3.41	3.37	3.35	3.33	3.31	3.30	3.29	3.28
	.005	3.88	3.85	3.81	3.77	3.75	3.73	3.71	3.70	3.69	3.68
	.001	4.66	4.62	4.58	4.54	4.52	4.50	4.48	4.47	4.46	4.45

(Continued)